

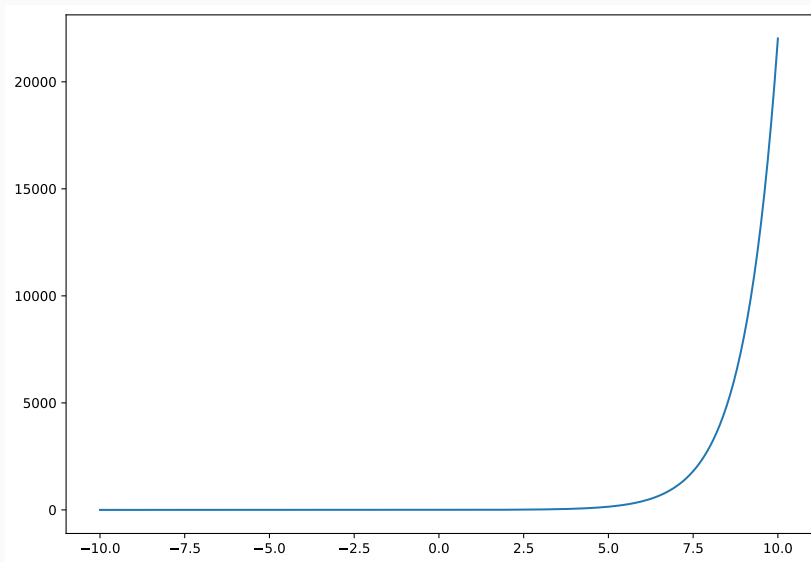
Lecture 10

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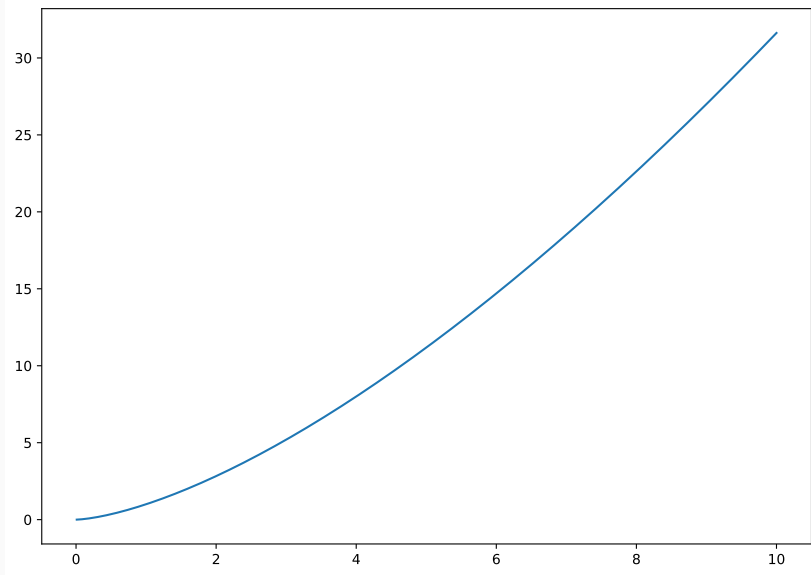
Examples of Convex Functions

Univariate convex functions

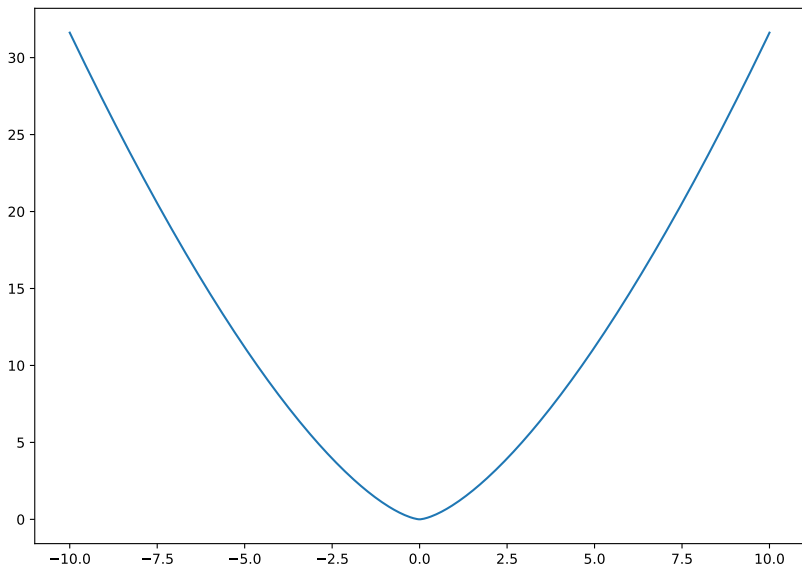
$f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = e^{ax+b}$



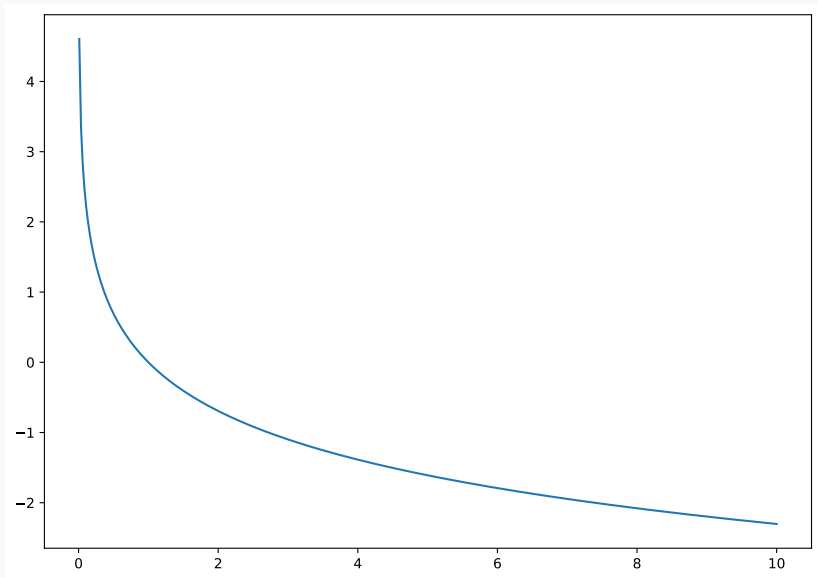
$f : \mathbb{R}_{++} \rightarrow \mathbb{R}$ given by $f(x) = x^p$, $p \leq 0$ or $p \geq 1$



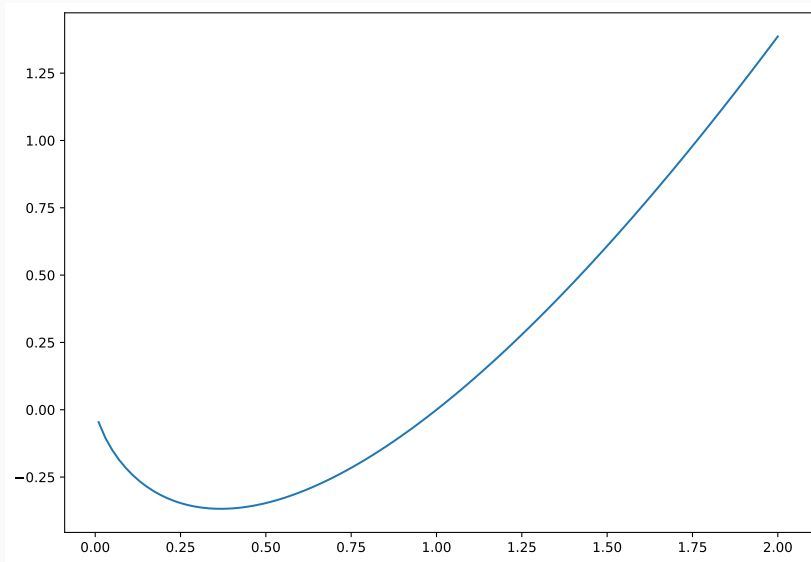
$f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = |x|^p$, $p \geq 1$



$f : \mathbb{R}_{++} \rightarrow \mathbb{R}$ given by $f(x) = -\log x$



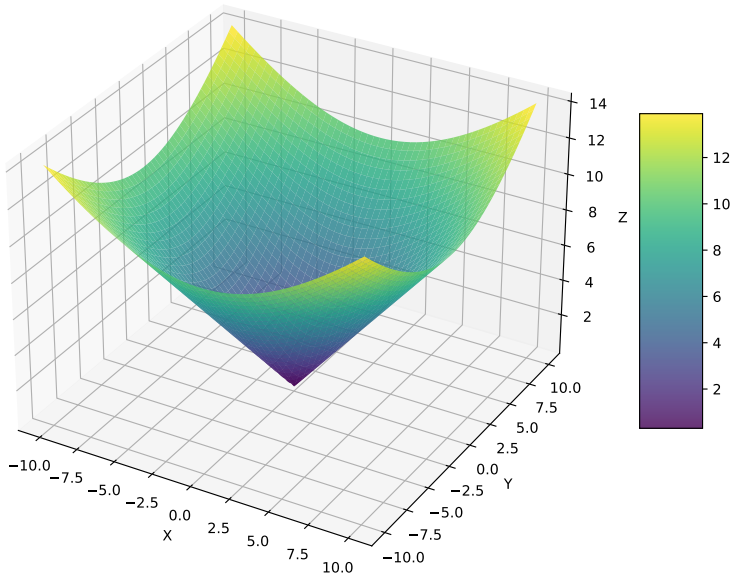
$f : \mathbb{R}_+ \rightarrow \mathbb{R}$ given by $f(x) = x \log x$ for $x > 0$ and $f(0) = 0$ (negative entropy)



Multivariate convex functions

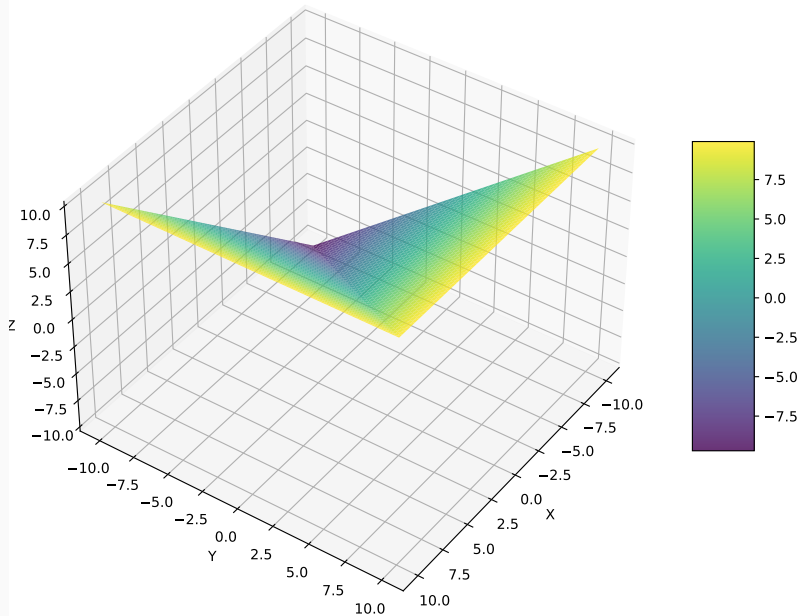
$\| \cdot \| : \mathbb{R}^n \rightarrow \mathbb{R}$ (for any norm)

3D Surface Plot of Vector Norm



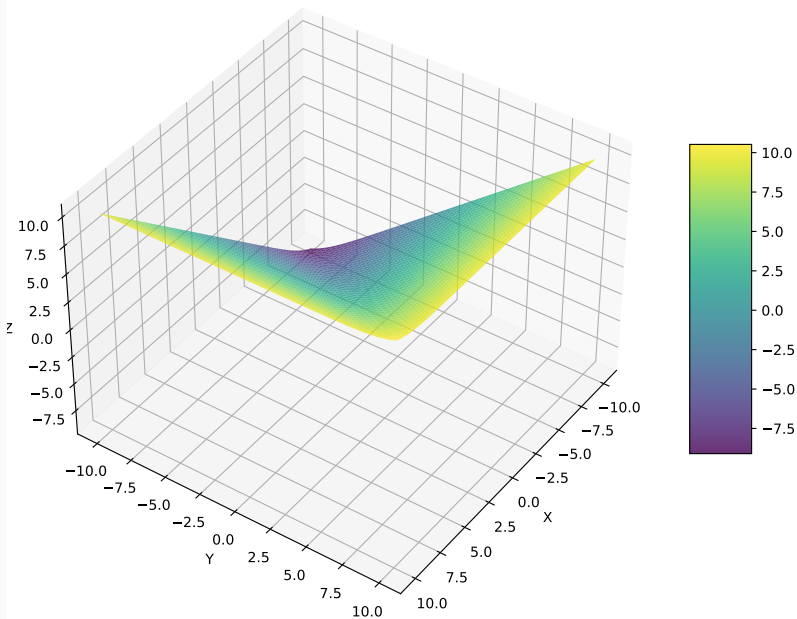
$\max : \mathbb{R}^n \rightarrow \mathbb{R}$ given by $\max(\mathbf{x}) = \max \{x_1, \dots, x_n\}$

3D Surface Plot of Maximum Function

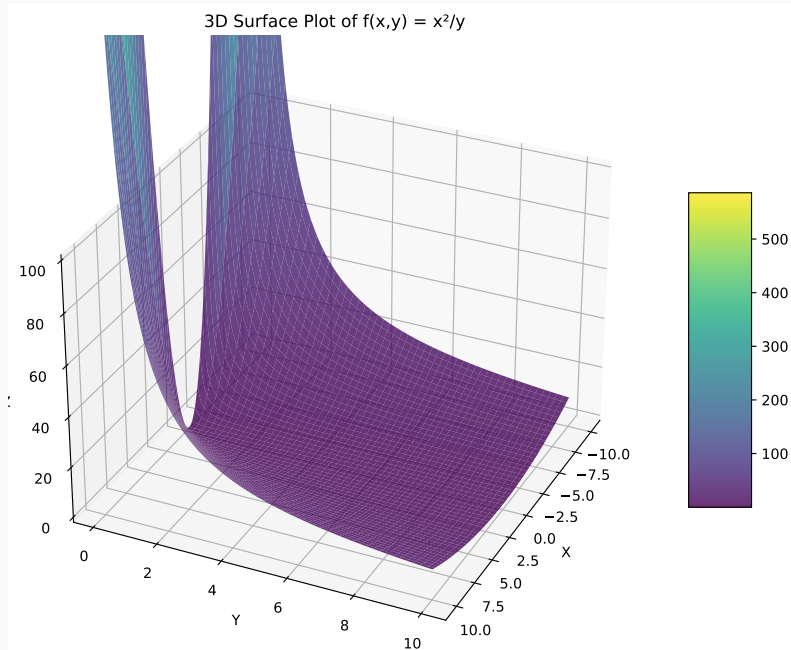


$\text{lse}: \mathbb{R}^n \rightarrow \mathbb{R}$ given by $\text{lse}(\mathbf{x}) = \log(e^{x_1} + \dots + e^{x_n})$

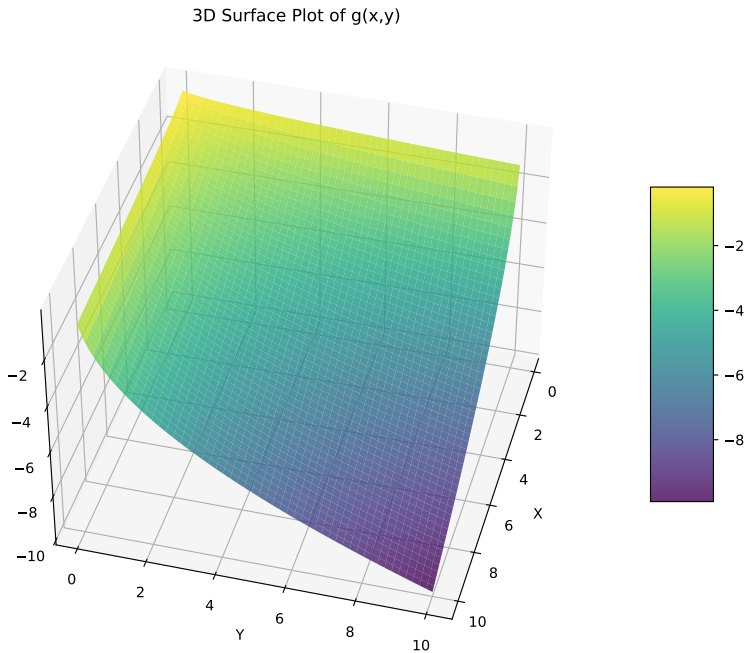
3D Surface Plot of $\text{lse}(x,y)$



$f : \mathbb{R} \times \mathbb{R}_{++} \rightarrow \mathbb{R}$ given by $f(x, y) = \frac{x^2}{y}$.



$g : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$ given by $g(\mathbf{x}) = -\sqrt[n]{x_1 x_2 \cdots x_n}$ (negative geometric mean)



$h : \mathbb{S}_{++}^n \rightarrow \mathbb{R}$ given by $h(X) = -\log \det X$

We can't visualize this one because it's a function of a matrix!

$\lambda_{\max} : \mathbb{S}_{++}^n \rightarrow \mathbb{R}$ by $X \mapsto \lambda_{\max}(X)$ (largest eigenvalue)

We also can't visualize this one...

Exercise

Prove that the functions in the examples above are convex.

The maximum function is continuous but not necessarily differentiable. The log-sum-exp function is also convex and serves as a “differentiable proxy” for a non-differentiable function because

$$\max(x_1, \dots, x_n) \leq \log(e^{x_1} + \dots + e^{x_n}) \leq \max(x_1, \dots, x_n) + \log n.$$

As an exercise, prove this inequality. Its gradient is often called softmax

$$\nabla \text{lse}(x_1, \dots, x_n) = \frac{1}{\sum_{i=1}^n e^{x_i}} \begin{bmatrix} e^{x_1} \\ \vdots \\ e^{x_n} \end{bmatrix} = \text{softmax}(x_1, \dots, x_n),$$

and is very important in reinforcement learning and in neural networks.

Note that $\text{softmax} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ - the coordinates of its output are all in $[0, 1]$ and sum to one, i.e., they may be interpreted as probability values, a very useful feature.

The epigraph condition

Another useful condition for checking convexity also relates convex functions to convex sets.

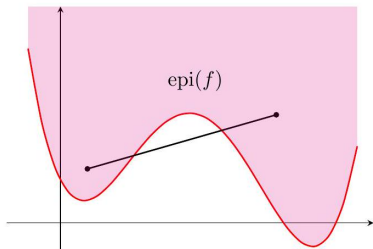
Definition (Epigraph)

The epigraph of a function $f : \Omega \rightarrow \mathbb{R}$ is the set of all points lying on or above the graph of f :

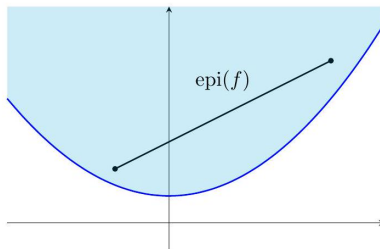
$$\text{epi}(f) = \{(\mathbf{x}, t) \in \Omega \times \mathbb{R} : f(x) \leq t\}$$

Theorem (Set theoretic condition for convexity)

A function $f : \Omega \rightarrow \mathbb{R}$ is convex if and only if the epigraph $\text{epi}(f)$ is convex.



(a) Non-convex epigraph



(b) Convex epigraph

This example is in your homework but here we will use the epigraph condition to establish convexity.

Example

Let $f : \mathbb{R}^n \times \mathbb{S}_{++}^n \rightarrow \mathbb{R}$ be given by $f(\mathbf{x}, Y) = \mathbf{x}^\top Y^{-1} \mathbf{x}$. (This function shows up in many applications!) Show that f is convex.

To see that f is convex, we check that the epigraph of f is convex:

$$\text{epi}(f) = \{(\mathbf{x}, Y, t) \in \mathbb{R}^n \times \mathbb{S}_{++}^n \times \mathbb{R} \mid \mathbf{x}^\top Y^{-1} \mathbf{x} \leq t\}.$$

Well, this is pretty messy. It's got the inverse matrix Y^{-1} in it.

We can simplify this by using the **Schur complement condition** for positive semidefiniteness of block matrices.

Definition (Schur complement)

For a block matrix of the form:

$$M = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix}$$

We define the “Schur complement of A in M ” as:

$$S_A = C - B^T A^{-1} B$$

Alternatively, we can define the Schur complement of C in M as:

$$S_C = A - B C^{-1} B^T$$

Theorem (Schur complement condition)

For a block matrix of the form:

$$M = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix},$$

If A is positive definite, then M is positive semidefinite if and only if:

$$S_A \succeq 0$$

(You can show this by doing a block Gaussian elimination.)

In our case, we have

$$\text{epi}(f) = \{(\mathbf{x}, Y, t) \in \mathbb{R}^n \times \mathbb{S}^{++}_n \times \mathbb{R} \mid \mathbf{x}^\top Y^{-1} \mathbf{x} \leq t\}$$

We are going to consider the block matrix M given by:

$$M = \begin{pmatrix} Y & \mathbf{x} \\ \mathbf{x}^\top & t \end{pmatrix}$$

The Schur complement of Y in M is given by:

$$S_Y = t - \mathbf{x}^\top Y^{-1} \mathbf{x}$$

In this case, S_Y is a scalar. So $S_Y \succeq 0$ just means that $S_Y \geq 0 \implies t - \mathbf{x}^\top Y^{-1} \mathbf{x} \geq 0$.

This is exactly the defining condition for $\text{epi}(f)$! It tells us that a pair $(\mathbf{x}, Y, t)$ is in the epigraph of f if and only if $S_Y \succeq 0$.

A pair $(\mathbf{x}, Y, t)$ is in the epigraph of f if and only if $S_Y \succeq 0$.

Remember, the Schur complement condition tells us that $S_Y \succeq 0$ if and only if $M \succeq 0$.
(Given that $Y \in \mathbb{S}^{++n}$, which it is by assumption.)

$$M = \begin{pmatrix} Y & \mathbf{x} \\ \mathbf{x}^\top & t \end{pmatrix}$$

So we can now say that a pair $(\mathbf{x}, Y, t)$ is in the epigraph of f if and only if $M \succeq 0$.

So we can rewrite the epigraph of $f(\mathbf{x}, Y) = \mathbf{x}^\top Y^{-1} \mathbf{x}$ as:

$$\text{epi}(f) = \left\{ (\mathbf{x}, Y, t) \in \mathbb{R}^n \times \mathbb{S}^{++n} \times \mathbb{R} \mid \begin{bmatrix} Y & \mathbf{x} \\ \mathbf{x}^\top & t \end{bmatrix} \succeq 0 \right\}$$

$$\text{epi}(f) = \left\{ (\mathbf{x}, Y, t) \in \mathbb{R}^n \times \mathbb{S}^{++n} \times \mathbb{R} \mid \begin{bmatrix} Y & \mathbf{x} \\ \mathbf{x}^\top & t \end{bmatrix} \succeq 0 \right\}$$

We want to show that the epigraph of $f(\mathbf{x}, Y) = \mathbf{x}^\top Y^{-1} \mathbf{x}$ is convex.

We know that the set of positive semidefinite matrices is convex. That is, the set of all matrices A such that $A \succeq 0$ is convex.

How can we use this to show that the set of all triples $(\mathbf{x}, Y, t)$, fulfilling the condition above, is convex?

We will use the fact that the **linear maps preserve convexity**.

$$\begin{aligned}\text{epi}(f) &= \left\{ (\mathbf{x}, Y, t) \in \mathbb{R}^n \times \mathbb{S}^{++^n} \times \mathbb{R} \mid \begin{bmatrix} Y & \mathbf{x} \\ \mathbf{x}^\top & t \end{bmatrix} \succeq 0 \right\} \\ &= \left\{ (\mathbf{x}, Y, t) \in \mathbb{R}^n \times \mathbb{S}^{++^n} \times \mathbb{R} \mid \begin{bmatrix} Y & \mathbf{x} \\ \mathbf{x}^\top & t \end{bmatrix} \in \mathbb{S}_+^{n+1} \right\}\end{aligned}$$

We can define a map $L : \mathbb{R}^n \times \mathbb{S}^{++^n} \times \mathbb{R} \rightarrow \mathbb{S}^{n+1}$ by:

$$L(\mathbf{x}, Y, t) = \begin{bmatrix} Y & \mathbf{x} \\ \mathbf{x}^T & t \end{bmatrix}$$

So our epigraph becomes:

$$\text{epi}(f) = \{ (\mathbf{x}, Y, t) \in \mathbb{R}^n \times \mathbb{S}^{++^n} \times \mathbb{R} \mid L(\mathbf{x}, Y, t) \in \mathbb{S}_+^{n+1} \}$$

$$\text{epi}(f) = \{(\mathbf{x}, Y, t) \in \mathbb{R}^n \times \mathbb{S}^{n+1}_+ \times \mathbb{R} \mid L(\mathbf{x}, Y, t) \in \mathbb{S}^{n+1}_+\}$$

Or, equivalently,

$$\text{epi}(f) = L^{-1}(\mathbb{S}^{n+1}_+)$$

$$\text{epi}(f) = L^{-1}(\mathbb{S}_+^{n+1})$$

Let's remember our original question: we want to show that the epigraph of $f(\mathbf{x}, Y) = \mathbf{x}^\top Y^{-1} \mathbf{x}$ is convex.

We know that the preimage of a convex set under a linear map is convex.

We can easily check that L is a linear map, and we know that $\mathbb{S}_+^{n+1}$ is convex.

Therefore, the epigraph of $f(\mathbf{x}, Y) = \mathbf{x}^\top Y^{-1} \mathbf{x}$ is convex.

(Whew!)

Example: log determinant function

Recall that the Hessian of the negative log determinant function $f(X) = -\log \det(X)$ is the linear map $\nabla^2 f(X) : \mathbb{S}^n \rightarrow \mathbb{S}^n$,

$$[\nabla^2 f(X)](H) = X^{-1} H X^{-1}$$

Can we show that the Hessian is positive semidefinite?

We will need to show that the inner product of any matrix H with the result of applying the Hessian operator to H is non-negative. (This is the same as saying that the Hessian operator is positive semidefinite.)

For matrices, remember that the inner product is given by $\text{tr}(H^\top K)$.

So we will want to show that

$$\text{tr}(H^\top \nabla^2 f(X) H) = \text{tr}(H^\top X^{-1} H X^{-1}) \geq 0$$

We will use two facts to simplify this expression:

1. Because X is positive definite, we can write $X = RR^\top$ for some invertible matrix R . Then $X^{-1} = R^{-\top}R^{-1}$.
2. Because H is a symmetric matrix, we have $H^\top = H$.

So we can rewrite the expression as:

$$\text{tr}(H^\top \nabla^2 f(X) H) = \text{tr}(H^\top X^{-1} H X^{-1}) = \text{tr}(H^\top R^{-\top} R^{-1} H R^{-\top} R^{-1})$$

Doing some algebra, we get:

$$\begin{aligned}\mathrm{tr}(H^\top \nabla^2 f(X) H) &= \mathrm{tr}(H^\top X^{-1} H X^{-1}) \\ &= \mathrm{tr}(R^{-1} H R^{-\top} R^{-1} H R^{-\top}) \\ &= \mathrm{tr}\left((R^{-1} H R^{-\top})^\top (R^{-1} H R^{-\top})\right) \\ &= \|R^{-1} H R^{-\top}\|^2 \geq 0;\end{aligned}$$

OK! We have shown that the Hessian is positive semidefinite, and so the negative log determinant function is convex.

Basic quantities in multivariable optimization

We begin with the basic definitions.

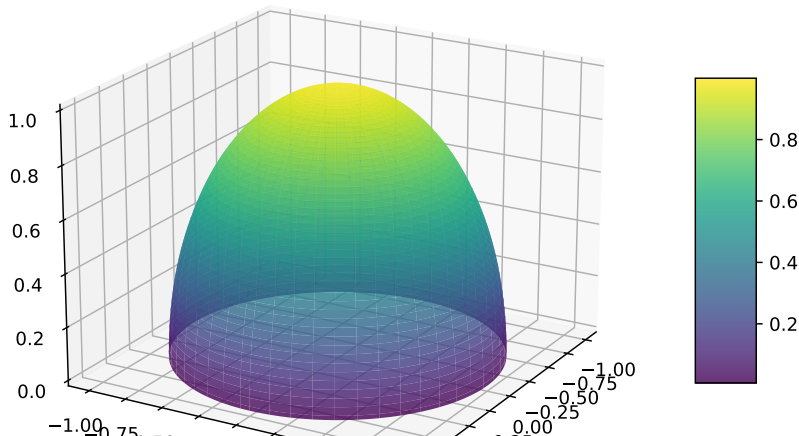
For a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the derivative at a point $\mathbf{x}$ is a linear map $Df(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^m$. This map transforms directions in the input space to directions in the output space.

Specifically, it maps from directions in the input space where we might make infinitesimal moves, to directions in the output space where the function value will infinitesimally change.

Example one: $f : \mathbb{R}^2 \rightarrow \mathbb{R}$

For $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, the derivative is a linear map from $\mathbb{R}^2$ to $\mathbb{R}$. It tells you that if you move an infinitesimal amount in the a given direction $\mathbf{h}$ in the x, y plane, the function changes by the (1-dimensional) scalar $Df(\mathbf{x}) \cdot \mathbf{h}$ in the z direction.

Half-Sphere Surface Plot



At most points, we can pick an infinitesimal direction $\mathbf{h}$ to move in which will change the function value in its one dimension (up or down).

The derivative maps the $x - y$ plane to the 1-dimensional subspace which represents possible directions of change along the z axis.

Because it maps to a 1-dimensional space, the derivative has rank 1.

But at some points, no matter which direction we move in, the function value doesn't change at all. The derivative always maps to the zero vector, which is a zero-dimensional subspace. At these points, the derivative has rank 0.

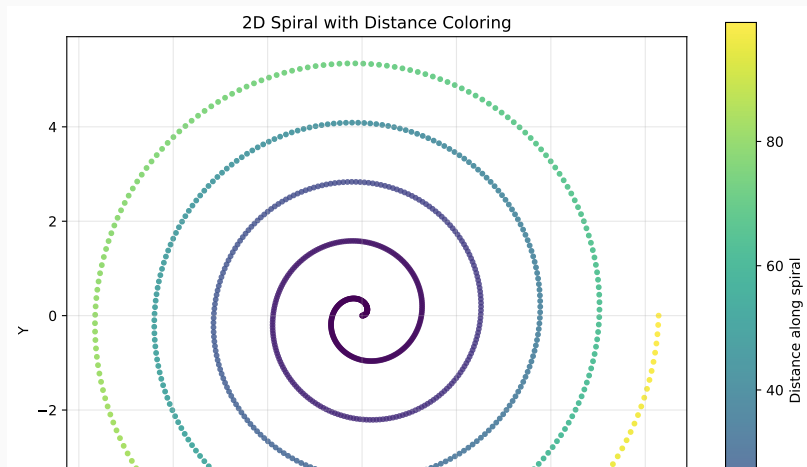
These are the critical points!

Example two: $f : \mathbb{R} \rightarrow \mathbb{R}^2$

By contrast, for $f : \mathbb{R} \rightarrow \mathbb{R}^2$, the derivative is a map from $\mathbb{R}$ to $\mathbb{R}^2$.

Suppose we have a function that traces out a spiral in the x, y plane:

$$f(t) = \begin{bmatrix} t \cos t \\ t \sin t \end{bmatrix}$$



The derivative of f at t is the linear map $Df(t) : \mathbb{R} \rightarrow \mathbb{R}^2$ given by

$$Df(t)(h) = \begin{bmatrix} \cos t - t \sin t \\ \sin t + t \cos t \end{bmatrix} h.$$

This maps from the 1-dimensional tangent space at t (representing infinitesimal perturbations in t) to the 2-dimensional space $\mathbb{R}^2$ of possible tangent vectors to the image.

However, at any point along the spiral, the **range** of $Df(t)$ is only a 1-dimensional subspace of $\mathbb{R}^2$.

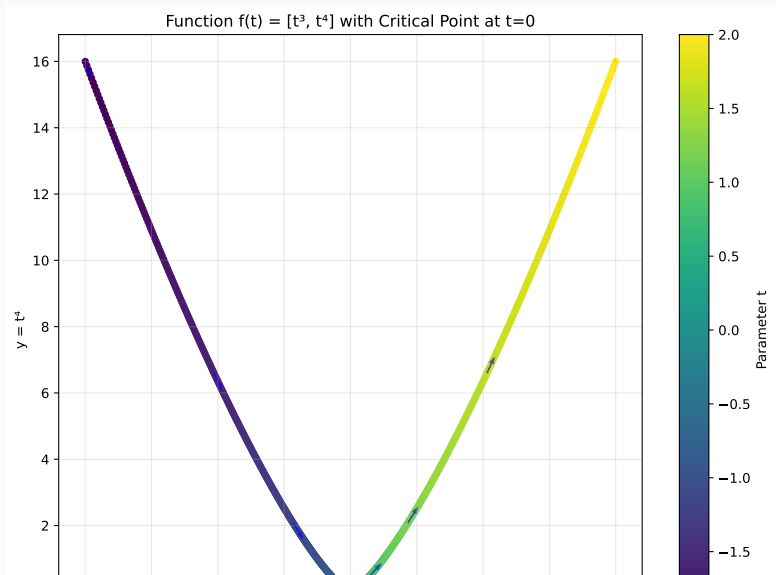
Infinitesimal changes in the parameter t can only produce directional changes tangent to the spiral, not in arbitrary directions in the x - y plane. Therefore, the derivative has rank 1.

So far, we have seen that the derivative can have *at most* the minimum of the dimensions of the input and output spaces. When the derivative achieves this maximum rank, we say that it is a full-rank linear map.

We have also seen that something special happens when the derivative has less than this maximum rank.

Let's look at another example of a function $f : \mathbb{R} \rightarrow \mathbb{R}^2$:

$$f(t) = \begin{bmatrix} t^3 \\ t^4 \end{bmatrix}$$



Let's look at the gradient of this function:

$$\nabla f(t) = \begin{bmatrix} 3t^2 \\ 4t^3 \end{bmatrix}$$

We see that the gradient is the zero vector at the origin.

At the origin, no matter which infinitesimal direction along the t axis we move in, we don't move at all in the $x - y$ plane. The rank of the derivative here is 0.

Example that we can't easily visualize

We can imagine a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that takes a point in the $x - y$ plane and returns a point in the $w - z$ plane.

Suppose we are at a point (x, y) , which corresponds to a point (w, z) in the $w - z$ plane.

We can ask ourselves: how many directions in the $w - z$ plane can we move in, just from making an infinitesimal change in the $x - y$ plane? (This is the rank of the derivative.)

Possible answers: 2, 1, 0. (draw each case, using two separate axes.)

If the rank is not maximal, we say that we are at a **critical** or **stationary** point.

If the rank is maximal, we say that we are at a **regular** point.

Definition (Stationary/critical point)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ or $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$ be differentiable. We say $\mathbf{x}_*$ or X_* is a stationary/critical point of f if

$$\text{rank } Df(\mathbf{x}_*) < m \quad \text{or} \quad \text{rank } Df(X_*) < pq.$$

In particular if $m = 1$ or $p = q = 1$, then we have a scalar-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ or $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ and in which case the condition above becomes

$$Df(\mathbf{x}_*) = 0 \quad \text{or} \quad Df(X_*) = 0$$

i.e., the derivative is the zero linear map (the linear map that sends every element to the zero vector/matrix). Alternatively $\mathbf{x}_*$ or X_* is a stationary/critical point of f if

$$\nabla f(\mathbf{x}_*) = 0 \quad \text{or} \quad \nabla f(X_*) = 0$$

in which case the value $f(\mathbf{x}_*)$ or $f(X_*)$ is called a stationary/critical value of f .

The inverse function theorem tells us that for functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, if the derivative has full rank, then the function is locally invertible.

Theorem (Inverse function theorem)

If (and only if) $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ has rank $Df(\mathbf{x}_) = n$, then f is invertible in an neighborhood of $\mathbf{x}_*$.*

Suppose we have a function $f(\mathbf{x}, \mathbf{y}) = \mathbf{z}$, where $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y}, \mathbf{z} \in \mathbb{R}^m$. If the derivative with respect to $\mathbf{y}$ has full rank at a given point, then we can create any possible small change in $\mathbf{z}$ just making small changes in $\mathbf{y}$ – we don't need to change $\mathbf{x}$ at all.

Alternatively, this means that we can keep $\mathbf{z}$ constant even while we change $\mathbf{x}$. All we have to do is find a small change in $\mathbf{y}$ that will counteract the effect of the small change in $\mathbf{x}$, and this will always be possible if the derivative has full rank.

This is the idea of an **implicit function**.

Theorem (Implicit function theorem)

If $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ has $\text{rank } D_{\mathbf{y}}f(\mathbf{x}_, \mathbf{y}_*) = m$, then the equation $f(\mathbf{x}, \mathbf{y}) = 0$ can be solved to give $\mathbf{y} = g(\mathbf{x})$ in an neighborhood of $\mathbf{x}_*$. Furthermore, $Dg(\mathbf{x}) = -[D_{\mathbf{y}}f(\mathbf{x}, g(\mathbf{x}))]^{-1} D_{\mathbf{x}}f(\mathbf{x}, g(\mathbf{x}))$ for all $\mathbf{x}$ in that neighborhood.*

Kinds of minimizers

A global minimizer is a point where the function value is lower than all other points in the domain.

It represents the best possible solution to a minimization problem.

Definition (Global minimizer)

Suppose $f : \Omega \rightarrow \mathbb{R}$, where $\Omega \subset \mathbb{R}^n$ or $\mathbb{R}^{m \times n}$ is open. We say $\mathbf{x}_* \in \Omega$ is a **global minimizer** if $f(\mathbf{x}_*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$.

Finding global minimizers is generally very difficult for non-convex functions, since there's no local characterization that guarantees global optimality.

A local minimizer is a point where the function value is lower than all other points in some neighborhood of the point.

Our formal definition will use an open ball of radius δ around the point.

Definition (Local minimizer)

Let f be as above. We say $\mathbf{x}_* \in \Omega$ is a **local minimizer** if $f(\mathbf{x}_*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in N \subset \Omega$, where N is an open subset (“neighborhood”) of Ω . Without loss of any generality, we can assume $N = B(\mathbf{x}_*, \delta)$ for some $\delta > 0$, so alternatively, $\mathbf{x}_* \in \Omega$ is local minimizer iff there exists $\delta > 0$ such that $f(\mathbf{x}_*) \leq f(\mathbf{x})$ for all $\mathbf{x}$ such that $\|\mathbf{x} - \mathbf{x}_*\| < \delta$.

A strict local minimizer is a local minimizer that is strictly lower than all other points in some neighborhood of the point. It means that there is no flat region around the point.

Definition (Strict local minimizer)

Let f be as above. We say $\mathbf{x}_* \in \Omega$ is a strict local minimizer if there is δ such that $f(\mathbf{x}_*) < f(\mathbf{x})$ for all $\mathbf{x}$ such that $0 < \|\mathbf{x}_* - \mathbf{x}\| < \delta$.

Optimization

Global optimization is very difficult and effectively impossible.

We shall instead look at local minimization problems, such as convex optimization problems.

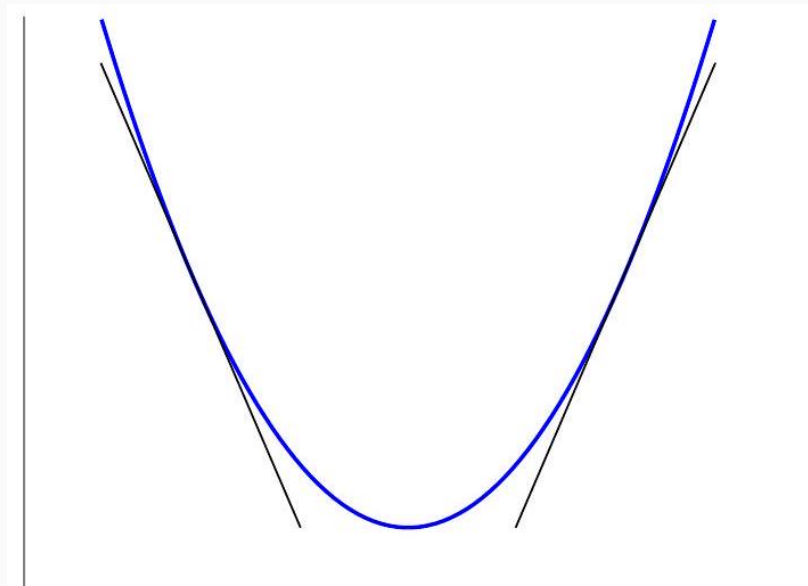
In local minimization problems, the goal is to find the local minimizer, not the local minimum (if we have the former, we can compute the latter).

But even local optimization is difficult if we don't assume something more about the function f ! The most common assumption is that $f \in C^1(\Omega)$ or $C^2(\Omega)$.

There are more specialized assumptions like $\nabla f \in \text{Lip}_\gamma(\Omega)$ that can be convenient in proofs.

Characterizing local minimizers in terms of derivatives

Remember, for the univariate case, if x is a local minimizer, then the derivative on the left and right of x must have opposite signs.



We said that for $f : \mathbb{R} \rightarrow \mathbb{R}$ differentiable at x , if x is a local minimizer, then $f'(x) = 0$. If x is the local minimizer, then for y sufficiently close to x , we have

$$\begin{aligned}\frac{f(y)-f(x)}{y-x} &\leq 0 && \text{for } y < x \\ \frac{f(y)-f(x)}{y-x} &\geq 0 && \text{for } y > x\end{aligned}$$

Thus the derivative either does not exist, or $f'(x) = 0$.

We will derive similar results in the multivariate case. In particular, we will give first and second order optimality conditions. We will find that the multivariate versions of theorems in univariate optimization are quite harder to prove.